

Federated Crime Prediction: Complete Project Handover

Everything from the original paper to the current state

Written in plain English so that any researcher — or another AI assistant — can pick this up with no prior context. Includes every approach tried, every formula, every result, and every failure.

Prepared by: Gaurav

Date: 07 August 2026

0. Read this first (one-page summary)

Where it started. Reproduce a research paper called **FedCrime** (Neurocomputing 679, 2026). It predicts city crime using *federated learning* (each city area trains on its own private data; only model numbers are shared) plus a *ZINB loss* that copes with the huge number of days when no crime happens.

Where it ended. After reproducing it we tried many extensions. Most were borrowed techniques or outright failures. But one line of work turned into a real, publishable finding:

KEY: Fairness gaps reported with F1 in sparse crime prediction are **mostly a measurement artifact**. The maximum possible F1 depends on how often crime happens, so quiet neighbourhoods can never score as high as busy ones — even with an equally good model. We prove this, verify it on 4 cities, re-analyse 7 published models, and show it **flips where remediation resources would be sent**.

Current status. The research is done and verified. The manuscript is not written yet. See Section 12 for exactly what remains.

Item	Status
Reproduce FedCrime	DONE — matches published numbers
Extensions (GNN, week-ahead, uncertainty, attacks, defenses)	DONE — mostly borrowed or failed
Base-rate / SkillGap discovery	DONE — the core contribution
4-city empirical study	DONE — LA, Chicago, NYC, SF
Literature audit (7 published models)	DONE
Decision-impact analysis	DONE — targeting flips 6/6
fairaudit package + theory extensions	DONE
Write the paper	NOT STARTED

1. The original paper (FedCrime) — what it does

Crime prediction = for each city area and each day, predict which of 8 crime types will occur. It is a *multi-label binary* problem: 8 yes/no answers per area per day.

Three problems FedCrime addresses:

- (a) **Privacy** — older methods pool all precinct data centrally. FedCrime keeps data local and shares only model weights.
- (b) **Zero-inflation** — most area-days have no crime (about 69% zeros in Los Angeles). Ordinary models collapse to always predicting 'nothing'.
- (c) **Long tail** — a few busy areas, many quiet ones.

1.1 The model: TCN-SD

A **Temporal Convolutional Network** reads the last 8 days, then a **Statistical Distribution head** outputs three numbers per crime type.

```
Eq 3: h(l,t) = ReLU( W1 . X(t - d*k) + b1 )      dilated causal convolution
Eq 4: h(l+1,t) = ReLU( W2 . h(l, t - d*k) + b2 )
Eq 5: h_tcn(t) = h(l+1,t) + Residual(X)           skip connection

Eq 6: pi = sigmoid(W_pi . h + b_pi)  zero-inflation probability, in [0,1]
Eq 7: mu = exp(W_mu . h + b_mu)      mean count, > 0
Eq 8: phi = softplus(W_phi . h + b_phi) dispersion (spread), > 0
```

In words: π = 'is this a structural no-crime day?'; μ = 'if crime happens, how much?'; ϕ = 'how bursty is it?'. Dilation $d = 1,2,4$ lets the network see far back cheaply. The skip connection keeps gradients flowing.

1.2 The ZINB loss (the paper's main idea)

```
Eq 9: L_zero = 1{Y=0} * log(pi + eps)
Eq 10: L_NB = 1{Y=1} * [ logGamma(Y+phi) - logGamma(phi) - logGamma(Y+1)
                        + phi*(log phi - log(phi+mu))
                        + Y*(log mu - log(phi+mu)) ]
Eq 11: J = - E[ L_zero + L_NB ]      (minimise)
```

The zero term rewards confident 'no crime' on quiet days; the NB term shapes the rare real events. Effect: sparse areas produce small, calm weight updates; busy areas produce sharp informative ones. That is why plain averaging works on the server.

1.3 Federated averaging

```
Eq 12: W = (1/|R|) * sum over clients i of w_i
```

The server simply averages client models. The paper's claim: fix the loss and you do not need a clever server.

1.4 The data

Dataset	Regions	Types	Sparsity	Source
Los Angeles 2018	113	8	68.7% zeros	FedCrime GitHub (bundled)
Chicago 2015	77	8	55.4%	City open-data portal
New York 2019	77 precincts	8	44.0%	NYC open data
San Francisco 2019	41	8	68.9%	SF open data
Chicago 311 (2019)	77	8	(2nd domain)	City open data

Split by TIME, never randomly: ~6.5 months train, 0.5 month validation, 1 month test. Window = 10 days (8 days input, predict the next).

2. Step 1 — reproducing FedCrime

We rebuilt the whole method from scratch in PyTorch and ran it on the real LA data.

Subset	Our Macro-F1	Paper's
SET-Alpha (easiest)	71.70	72.16
SET-Beta	65.18	61.70
SET-Gamma	55.81	48.54
SET-Omega (hardest)	49.42	48.43

Subsets are built by ranking areas by crime and splitting at the 20/30/50 percentiles into **Head** (busy 20%), **Mid** (30%), **Tail** (quiet 50%). Each subset mixes them differently, so sparsity rises Alpha to Omega.

HONEST NOTE: One fix was needed: on real data the model first collapsed to predicting 'no crime' everywhere ($F1 \sim 3$). We added class weighting (rare crimes weighted up to 6.8x) plus a classification head, which fixed it.

```
loss = BCE_with_pos_weight(logit, Y) + 0.3 * ZINB_loss
pos_weight_c = (#negatives_c / #positives_c), clipped to [1, 10]
```

3. Step 2 — adding a graph (GNN), and the over-smoothing problem

FedCrime deliberately has **no graph**: areas never learn from neighbours. We added one. The real data has no ready-made graph, so we built it by connecting each area to its 6 most similar areas (correlation of daily crime).

```
corr = correlation between regions' daily crime series
A[i,j] = 1 if j is in region i's top-k most similar (+ self loops)
A_hat = D^-1/2 A D^-1/2 (normalised adjacency)
```

3.1 What went wrong: over-smoothing

Plain graph convolution: $H' = \text{ReLU}(A_hat \cdot H \cdot W)$

This averages every area with its neighbours. Busy and quiet areas get blended, blurring the signal. On real data the plain GNN scored **below** the no-graph baseline.

3.2 The two fixes (both standard, existing methods)

```
GATED (Li et al. 2016 style):
g = sigmoid(Wg . H) # a learned dial in [0,1] per region
H' = g * (A_hat . H . W) + (1-g) * H
-> busy regions turn the graph DOWN, quiet regions turn it UP

ATTENTION (GAT, Velickovic et al. 2018):
a_ij = softmax_j( (W h_i . W h_j) / sqrt(H) ) over graph neighbours
H'_i = sum_j a_ij * (W h_j)
-> learns HOW MUCH to trust each neighbour instead of averaging equally
```

City	Plain GNN	Gated	Attention	No-GNN
Los Angeles	48.09	53.94	53.21	54.09
Chicago	55.95	66.55	—	66.94

KEY: The gated/attention fixes recover the lost accuracy, but **never beat** the no-graph baseline. So in this domain the graph does not help accuracy at all — it only helps fairness (attention raised the Tail score from 24.79 to 28.04).

HONEST NOTE: Gated GNNs (2016) and GAT (2018) are well-known existing methods. We applied them; we did not invent them. Claiming novelty here would be rejected.

4. Step 3 — other extensions we tried

4.1 Week-ahead prediction

$X = \text{days } t \dots t+L-1 \quad \rightarrow \quad Y = \text{day } t + L + H - 1 \quad (H = 7 \text{ for a week})$

Result: 48.62 at 7 days vs 48.76 at 1 day; AUC 0.742. Almost no degradation.

KEY: This non-degradation is itself informative: it suggests the model learns **periodic and base-rate structure**, not short-term dynamics — which questions how much the complex temporal machinery contributes.

4.2 Uncertainty (MC-dropout)

Keep dropout ON at test time, run T forward passes,
uncertainty = standard deviation of the T predicted probabilities

Works, gives per-prediction confidence. Standard method.

4.3 Attacks by malicious clients

Attack	What it does	Formula
Label-flip	Trains on flipped labels	$Y \rightarrow 1 - Y$
Scaling	Blows up its update to dominate	$\text{delta} \rightarrow 8 * \text{delta}$
Sparsity-camouflage (ours)	Disguises poison as a quiet region	$Y \rightarrow 0; \text{delta} \rightarrow 0.3 * \text{delta}$

Finding: plain FedAvg **collapses** under the scaling attack (7.26 ± 14.52 on LA; AUC 0.470 on Chicago — worse than guessing), while every robust defense holds (~ 48.8 LA, ~ 62 Chicago). This is real but is a known result.

4.4 Our own defenses (strust, strust2) — FAILED

strust2: 1) clip each update to the MEDIAN norm (kills scaling attacks)
2) $\text{trust}_i = \max(0, \cos(\text{update}_i, \text{coordinate-median direction}))^2$
3) aggregate weighted by trust

HONEST NOTE: This failed. Across six tests $p = 0.12$ to 0.55 — never significantly better than existing defenses, and **significantly worse** than plain FedAvg on label-flip ($p = 0.008$). We report this as a negative result. Do not claim a new defense in the paper.

5. Step 4 — the fairness problem, and three failed fixes

The model scored ~69 on busy (Head) areas but only ~28 on quiet (Tail) areas — a 41-point gap. That looked like unfairness to poor neighbourhoods. We tried three fixes.

Attempt	Method	Result
1. Reweighting	Weight tail regions 2.3x in the loss	FAILED — gap got WORSE (Tail 19 -> 11)
2. Group-DRO	Adaptively focus on the worst group: w_g proportional to $\exp(\tau * \text{Loss}_g)$	FAILED — gap stuck at ~41 for $\tau = 3, 6, 10$
3. Adaptive thresholds	Tune a separate decision threshold per region x crime type on a validation set	WORKED — Tail 10-19 -> ~28, gap 55 -> ~41

KEY: Attempt 3 helped because sparse regions rarely cross a fixed 0.5 cutoff, so they predict nothing. Lowering their threshold lets them be predicted at all.

HONEST NOTE: But the gap would not fall below ~41 no matter what we did. Three different, well-founded methods all hit the same wall. That was the clue that something deeper was going on.

6. THE DISCOVERY — the gap is a measurement artifact

The reason nothing worked: **we were measuring with a broken ruler.**

6.1 The mathematics

If you simply predict 'crime happens' every single day, then for a label that is positive a fraction p of the time:

$$\text{precision} = p, \quad \text{recall} = 1, \quad \text{so} \quad F1 = 2p / (1 + p)$$

This is $A(p)$, the **ATTAINABILITY**: what a model with NO SKILL AT ALL scores.

$$A(p) \text{ increases with } p: \quad d/dp [2p/(1+p)] = 2/(1+p)^2 > 0$$

So a busy area ($p = 0.56$) has $A = 71.9$, while a quiet area ($p = 0.16$) has $A = 27.6$. **The quiet area cannot score as high, however good the model is.** Comparing raw F1 between them is like comparing marks from a test out of 100 with a test out of 40.

6.2 The formal framework

DEFINITION 1 $A_M(p)$ = value of metric M for a skill-free predictor at base rate p
 DEFINITION 2 $\text{Skill} = M(g) / A_M(p_g)$ (fraction of attainable achieved)

PROPOSITION 1 (decomposition)

$$\text{Gap}_M = [A_M(p_H) - A_M(p_T)] \quad \leftarrow \text{ATTAINABILITY term} \\
+ [A_M(p_H)(\text{Sk}_H - 1) - A_M(p_T)(\text{Sk}_T - 1)] \quad \leftarrow \text{SKILL term}$$

Even with EQUAL skill, the gap is nonzero if attainability depends on p .

PROPOSITION 2 $A_{F1}(p) = 2p/(1+p)$, strictly increasing -> F1 IS confounded

PROPOSITION 3 For a skill-free predictor $\text{AUC} = 0.5$ and balanced accuracy = 0.5
 for EVERY p -> these are INVARIANT, so their gaps measure skill

6.3 The metric taxonomy

CONFOUNDED (do not use alone)	INVARIANT (valid)
F1, precision, accuracy, average precision, demographic parity, predictive parity	AUC-ROC, balanced accuracy, equal opportunity, equalized odds

HONEST NOTE: Recall/TPR alone is attainability-invariant but **operating-point dependent** — it shifts if thresholds differ per group. Report it paired with FPR, or fix the operating point.

6.4 The corrected metric

Ratio form : Skill = $M / A(p)$ ('% of attainable')
unstable when A is small

SKILL SCORE : SkillScore = $(M - A) / (1 - A)$ <- RECOMMENDED
0 = no better than skill-free, 1 = perfect
bounded and comparable across very different base rates

7. The evidence

7.1 Controlled proof (skill identical by construction)

We simulated two groups with **exactly the same model skill**, differing only in base rate (0.56 vs 0.16):

Metric	Head	Tail	GAP	Class
F1	77.72	39.03	38.69	confounded
Precision	69.27	25.03	44.24	confounded
Accuracy	71.69	56.10	15.58	confounded
AUC-ROC	80.47	80.28	0.19	invariant
Balanced acc.	69.48	69.30	0.18	invariant

KEY: Skill is provably equal, yet F1 reports 38.7 points of 'unfairness' and AUC reports 0.19 — a 200x discrepancy on identical predictions.

7.2 Four real cities

City	F1 gap	AUC gap	Skill-F1 gap
Los Angeles	40.91	-4.77	-6.30
Chicago	44.91	-2.15	-1.27
New York	18.64	-2.38	-1.49
San Francisco	45.96	+3.83	-10.84

KEY: New York is the natural control. Its Head and Tail base rates are close (0.695 vs 0.475), and its F1 gap collapses to 18.64 — less than half the others — while its AUC gap is unchanged. Nothing about the model changed; only the base rates.

7.3 The gap is predictable with no model at all

City	Predicted by base rates alone	Trained model's gap	Error
LA	42.8	41.01	1.79
Chicago	45.9	44.47	1.43
NYC	19.9	18.55	1.35
SF	48.8	45.48	3.32

KEY: Correlation = **0.998**, mean error under 2 F1 points. A formula using no model predicts the trained model's 'unfairness'.

8. Re-analysis of published work

8.1 FedCrime's own published numbers

Using only the paper's Table 1 (per-category sparsity) and Tables 2/3/5 (scores) — recomputable by anyone:

Subset	Reported F1 (LA)	Skill-free ceiling	Skill
S-Alpha	72.16	71.2	101%
S-Beta	61.70	61.2	101%
S-Gamma	48.54	48.5	100%
S-Omega	48.43	45.8	106%

KEY: Correlation between published scores and the skill-free ceiling = **0.998**; mean difference 1.07 F1 points; mean skill 99.5%. And **107%** of the paper's headline 'degradation under sparsity' is explained by the ceiling falling, not by lost skill.

8.2 Seven published models (benchmark paper arXiv:2407.19324)

That benchmark evaluates DeepCrime, MiST, CrimeForecaster, HAGEN, ST-HSL and AIST on Chicago, grouped by community size. It states the trend comes from 'more data -> better training'.

Test	Result
Models with $r > 0.8$ between macro-F1 and crime volume	4 of 6
Mean correlation, micro-F1 vs crime volume	+0.90
The one model that does NOT correlate	ST-HSL ($r = -0.43$)

KEY: Internal validation: the benchmark itself describes ST-HSL as using self-supervision that 'makes the model invariant to area or density'. The one model *designed* to be density-invariant is exactly the one that breaks the correlation — consistent with the base-rate explanation.

HONEST NOTE: This is correlation, not proof. Better learning and higher attainability predict the same direction. The honest claim is that the published evidence **cannot separate them**, because no invariant metric (e.g. AUC) is reported. Also: relative rankings between methods remain VALID, since the ceiling is shared within a group.

9. Does it change any real decision?

Decision	Changes?	Detail
D1. Which MODEL is best	NO — 0 of 5 groups	Ceiling is shared within a group, so ranking is preserved
D2. Which GROUP is worst served (where remediation goes)	YES — 6 of 6 models	Rank correlation between the two criteria = -0.70

KEY: The confound does **not** mislead you about which method to use. It misleads you about **where the model is failing** — which is precisely the fairness question. Raw F1 sends help to the sparsest regions almost by construction; the corrected measure points elsewhere in every model tested.

10. Full scoreboard — what worked, what did not, what is ours

#	What we tried	Worked?	Novel?
1	Gated / attention GNN (fix over-smoothing)	YES — 48->54 LA, 56->67 CHI	NO — GAT/gated GNN are standard
2	7-day-ahead prediction	YES — 48.62 vs 48.76	NO — routine; the non-degradation is interesting though
3	Uncertainty (MC-dropout)	YES	NO — standard
4	Sparsity-camouflaged attack	YES — damages FedAvg	MILD — poisoning is well studied
5	strust / strust2 defense (ours)	NO — $p = 0.12-0.55$	NO — failed to beat baselines
6	Fairness via reweighting	NO — made it worse	—
7	Fairness via Group-DRO	NO — stuck at ~41	—
8	Adaptive thresholds per region x type	YES — Tail 10-19 -> 28	MILD — tuning is standard
9	Base-rate ceiling / SkillGap	YES — best finding	YES — measurement correction
10	Trilemma	DISSOLVED, not solved	YES — a legitimate negative result
11	Skill vs trivial baseline	YES — AUC 0.74/0.75	NO — essential validation
12	Multi-seed statistics	YES — 5 seeds, t-tests	NO — basic rigor
13	Literature audit (7 models)	YES	YES — the checkable evidence
14	Fairness-metric taxonomy	YES	MILD — connects to Chouldechova
15	Second domain (Chicago 311)	YES	MILD — generalisation
16	fairaudit package	YES	MILD — a usable artifact
17	Decision-impact analysis	YES — targeting flips 6/6	YES — the 'so what'
18	Theory extended (multi-label/ranking/regression)	YES	MILD — extends known results
19	Skill-score fix	YES — stable +4.4 vs unstable -34	NO — standard forecasting form

KEY: The five real contributions: #9, #10, #13, #17, #18 — plus the #16 artifact. Everything else is either borrowed engineering or an honest failure.

11. Prior work you MUST cite (non-negotiable)

Our theory builds on established results. Failing to cite these would be an instant rejection.

Work	What it already established
Davis & Goadrich, ICML 2006	ROC is invariant to class skew; Precision-Recall is not. Defines the 'achievable PR curve'.
Boyd et al., 2012	The unachievable region in PR space determined by the base rate — very close to our attainability.
Chouldechova, 2017	Fairness criteria cannot be jointly satisfied when base rates differ across groups — in recidivism prediction.
Velickovic et al., 2018 (GAT)	Graph attention networks.
Li et al., 2016	Gated graph neural networks.

HONEST NOTE: Our contribution is NOT the metric property (known). It is: (a) showing it **dominates** reported gaps in this literature, (b) the four-city + 7-model empirical audit, (c) the decision-impact consequence, and (d) a corrected protocol and tool.

12. Files, how to run, and what remains

12.1 File inventory

File	Purpose
robust_fair_gnn.py	Main pipeline: data, ST-GNN, ZINB, attacks, defenses, metrics, multi-seed stats
theory.py	Propositions 1-3 with numerical verification
fairaudit/ (package)	metrics.py (taxonomy), extensions.py (multi-label/ranking/regression), audit.py (API)
baserate_analysis.py	Artifact-gap table across cities (no model needed)
reanalysis_published.py	Re-analysis of FedCrime's published tables
literature_audit.py	Audit of 7 published models
decision_impact.py	D1 model selection / D2 remediation targeting
scripts/preprocess_*.py	Build Chicago, NYC, SF, and Chicago-311 datasets
FedCrime_FullStudy.ipynb	Runs everything end to end (Colab/Kaggle, GPU)
fedcrime/ (package)	The original faithful FedCrime reproduction

12.2 How to run

```
# everything, in one notebook (recommended)
open FedCrime_FullStudy.ipynb in Colab -> Runtime -> GPU -> Run all

# or individual pieces
python theory.py                # the theory, verified
python -m fairaudit.extensions  # multi-label / ranking / regression
python literature_audit.py      # 7 published models
python reanalysis_published.py  # FedCrime's own tables
python decision_impact.py       # does the decision change?
python baserate_analysis.py     # artifact gap per city
python robust_fair_gnn.py --city la --gnn gated --adathr --metrics --seeds 3
```

12.3 What still has to be done

Task	Why
WRITE THE MANUSCRIPT	Results exist; the paper does not
Cite Davis-Goadrich, Boyd, Chouldechova	Blocking — reviewers will know these
Ethics / limitations section	Mandatory for predictive policing
Run the full study once, capture real outputs	Components verified separately; need one clean end-to-end run
Increase to 10 seeds on headline claims	Stronger statistics
Add professor as co-author	Framing and venue choice

12.4 Honest positioning for the paper

KEY: Say: 'Reported F1 fairness gaps in sparse crime prediction are largely attributable to base-rate differences rather than model bias; we prove this, verify it on four cities and seven published models, and show it changes where remediation is targeted.'

HONEST NOTE: Do NOT say: 'we invented a new defense' (six p-values disprove it), 'we discovered that F1 depends on base rate' (known since 2006), or 'we solved the trilemma' (it dissolved rather than being solved). Also always state that tail regions genuinely do have low absolute performance — the correction concerns attribution, not a claim that poor areas are well served.

Realistic venues: fairness venues, urban-computing journals, IEEE Access (strong fit); IEEE TKDE / ACM TIST as an experimental-and-analysis paper (reachable). Not top-tier ML conferences.

End of handover. Every number in this document was produced by the scripts listed in Section 12 and can be regenerated.